

HOTEL BOOKING DATA ANALYSIS

Project Documentation

From raw booking records to a governed Power BI dashboard — data cleaning, exploratory analysis, feature engineering and business recommendations.

Prepared by: Hamza Ahmed Bayoumi

Dataset: Hotel Bookings (City Hotel & Resort Hotel, Portugal) — 119,390 records, July 2015 – August 2017

Table of Contents

1. Project Overview and Business Objective	5
1.1 Business Context	5
1.2 Core Business Questions	5
1.3 Headline Scale	5
1.4 Project Deliverables.....	5
2. Dataset Overview	6
2.1 Dataset Size Through the Pipeline	6
2.2 Main Columns.....	6
Booking details	6
Stay details	6
Guest profile.....	6
Commercial fields.....	6
The target variable	6
3. Data Cleaning and Preprocessing.....	7
3.1 Handling Duplicates.....	7
A data-integrity issue caught during the process.....	7
3.2 Handling Missing Values.....	7
3.2.1 Decisions Made for Missing Values in Company.....	7
3.2.2 Decisions Made for Missing Values in Agent	8
3.2.3 Country and Children	8
3.3 Handling Invalid or Unusual Values.....	8
3.4 Outlier Detection and Treatment.....	8
3.4.1 Method: Screen Statistically, Decide Commercially.....	8
3.4.2 Why a Blind IQR Rule Would Have Failed on This Dataset.....	9
3.4.3 Column-by-Column Decisions	9
ADR (Average Daily Rate)	9
Party size (adults, children, babies).....	10
Stay length (stays_in_week_nights / stays_in_weekend_nights).....	10
Lead time	10
Zero-inflated behavioral counters.....	11
3.4.4 Outlier Decision Register and Applied Treatment	11
4. Exploratory Data Analysis (EDA).....	12
4.1 Dataset Overview After Cleaning	12
4.2 Statistical Analysis	12
4.3 Categorical and Numerical Analysis	13

Categorical composition.....	13
Numeric distributions.....	13
4.4 Bivariate Analysis — What Predicts a Cancellation.....	13
The counter-intuitive deposit finding.....	14
Channel reliability.....	14
Special requests — the cleanest protective signal.....	14
Lead time — the strongest continuous driver	14
The two hotels are different businesses	14
4.5 Reserved vs. Assigned Room — and a Target-Leakage Warning	14
4.6 Multivariate Analysis — The Hidden Guest-Origin × Channel Interaction.....	15
4.7 Correlation Analysis.....	15
5. Feature Engineering	17
5.1 Stay & Value Features	17
5.2 Temporal Features	17
5.3 Behavioral and Commercial Features	18
5.4 Data-Integrity Flags	19
5.5 Composite Cancellation Risk Score	19
5.6 Rejected Features and the Leakage Exclusion List.....	20
6. Data Visualization	21
6.1 Distribution Profiling	21
6.2 Categorical Composition Charts	21
6.3 Cancellation-Rate Bar Charts (Bivariate).....	21
6.4 Trend and Seasonality Charts.....	21
6.5 Heatmaps (Multivariate)	21
6.6 Correlation Heatmap and Driver Ranking	21
6.7 Risk-Tier and Revenue-Exposure Charts	22
7. Power BI Dashboard	23
7.1 Page 1 — Executive Overview	23
7.2 Page 2 — Customer Analysis	23
7.3 Page 3 — Cancellation Analysis.....	23
7.4 Page 4 — Revenue & Room Analysis.....	23
7.5 Filters and Slicers.....	23
7.6 Main Business Insights Highlighted by the Dashboard	23
8. Key Findings and Business Problems.....	25
8.1 Revenue at Risk — Reframing the Cancellation Problem	25
8.2 Ranked Drivers of Cancellation	25
8.3 Answers to the Core Business Questions.....	26

Which customers cancel the most?	26
Which booking channels are the most reliable?	26
Which customer types generate the highest ADR?	26
Which hotel performs better?	26
Which countries have the highest cancellation rate?	26
8.4 Main Problems Identified and Likely Reasons.....	26
8.5 Business Recommendations.....	27
9. Project Workflow.....	28
Stage 0 — Data Integrity Setup	28
Stage 1 — Outlier Analysis	28
Stage 2 — Exploratory Data Analysis	28
Stage 3 — Feature Engineering	28
Stage 4 — Business Insights	28
Stage 5 — Power BI Delivery	28
Summary of Outputs by Stage	28
Stated Limitations	29
10. AI Assistance.....	30
Conclusion	31

1. Project Overview and Business Objective

1.1 Business Context

Online travel agencies (OTAs) now generate the majority of hotel bookings, but a growing share of those bookings never arrive. This project analyzes 119,390 hotel reservation records spanning a City Hotel and a Resort Hotel in Portugal to understand booking behavior, quantify the revenue impact of cancellations, and turn the findings into an interactive Power BI dashboard for ongoing monitoring.

1.2 Core Business Questions

- Which guests and channels cancel the most, and why?
- How much booked revenue is genuinely at risk from cancellations?
- Which bookings should be flagged and managed proactively?
- How can the findings be delivered as a self-serve BI dashboard for non-technical stakeholders?

1.3 Headline Scale

Metric	Value
Total bookings analyzed (post-cleaning)	86,638
Overall cancellation rate	27.5% (27.68% pre-rounding)
Total booked revenue	€34.46M
Average Daily Rate (ADR)	106.34
Average stay length	3.63 nights

1.4 Project Deliverables

- A fully documented Python (pandas/NumPy/seaborn/matplotlib) data-cleaning and feature-engineering pipeline, structured as a four-part analytical notebook: Outlier Analysis, Exploratory Data Analysis, Feature Engineering, and Business Insights.
- A 4-page interactive Power BI dashboard (Executive, Customer, Cancellation, and Revenue & Room views), filterable by hotel, month, customer type and country.
- A quantified set of business recommendations tied directly to measured effects in the data.

Analytical approach

The project deliberately favors business-driven, evidence-based decisions over generic, automated data-cleaning rules. Every cleaning, outlier, and feature-engineering decision is justified against a specific pattern found in the data, and documented in a decision register so the reasoning is auditable.

2. Dataset Overview

The dataset is a public hotel-bookings dataset covering two properties in Portugal — a City Hotel and a Resort Hotel — with arrivals from July 2015 to August 2017. Only calendar year 2016 is fully represented; 2015 covers July–December only and 2017 covers January–August only, which is an important caveat for any year-over-year comparison.

2.1 Dataset Size Through the Pipeline

Stage	Rows	Columns	Notes
Raw file loaded	119,390	32	As downloaded
After removing exact duplicates	87,396	32	31,994 duplicate rows removed (26.8% of the raw file)
After business-driven outlier treatment	86,638	32 (+ working columns)	758 further rows removed as provable non-stays
After feature engineering	86,638	67	35 new engineered columns added

2.2 Main Columns

The 32 raw columns fall into four functional groups:

Booking details

hotel, lead_time, arrival_date_year / _month / _week_number / _day_of_month, market_segment, distribution_channel, deposit_type.

Stay details

stays_in_week_nights, stays_in_weekend_nights, reserved_room_type, assigned_room_type, meal.

Guest profile

adults, children, babies, country, is_repeated_guest, customer_type.

Commercial fields

adr (Average Daily Rate), total_of_special_requests, booking_changes, agent, company, previous_cancellations, previous_bookings_not_canceled, days_in_waiting_list, required_car_parking_spaces, reservation_status, reservation_status_date.

The target variable

is_canceled — a binary flag (1 = the booking was cancelled or was a no-show, 0 = the guest stayed). This is the variable every downstream rate, chart, and recommendation is measured against.

3. Data Cleaning and Preprocessing

3.1 Handling Duplicates

The raw file contains 31,994 exact duplicate rows (26.8% of the data). These were identified with `data.duplicated()` and removed with `data.drop_duplicates().reset_index(drop=True)`.

Why this is the single biggest correction in the project

Because the dataset has no booking ID, duplicates cannot be distinguished from genuinely repeated same-day group bookings with 100% certainty. Left in place, the duplicate rows inflate the cancellation rate to 37.0%; after removal, the true rate is 27.5% — a roughly 9.5 percentage-point distortion that would have propagated into every subsequent chart and conclusion. This assumption is flagged explicitly in the notebook's Limitations section for confirmation against the source system.

A data-integrity issue caught during the process

An earlier reload of the raw CSV file (executed after the de-duplication step, based on the notebook's execution counters) silently re-introduced the 31,994 duplicate rows before the missing-value imputations were run. The notebook explicitly re-validates data integrity with an idempotent guard cell that re-checks for duplicates and missing values before any further analysis proceeds:

```
if data.duplicated().sum() > 0:      data = data.drop_duplicates().reset_index(drop=True)  assert
data.duplicated().sum() == 0 and data.isnull().sum().sum() == 0
```

Because the imputation rules are row-wise and deterministic, de-duplicating after imputation produces a frame identical to de-duplicating before it (verified byte-identical at 87,396 rows), so no analytical work was lost by catching and fixing this out-of-order execution.

3.2 Handling Missing Values

Column	Missing Values	Treatment Rule
company	82,137 (94%)	Corporate segment + missing → "Unknown Company"; all other segments → "No Company"
agent	12,193	Direct segment + Direct channel + missing → "Direct Booking"; all other cases → "Unknown Agent"
country	452	Filled as "Unknown" — no reliable inference available from other fields
children	4	Filled with 0 (a count-type field; missing plausibly means none)

3.2.1 Decisions Made for Missing Values in Company

Rather than a blind imputation (mode-filling or row deletion), the missingness in company was analyzed against `market_segment` using a cross-tabulation:

```
pd.crosstab(data['market_segment'], data['company'].isna(), margins=True)
```

- Most bookings from Direct, Online TA, Offline TA/TO, Groups, Complementary and Aviation segments have no company ID at all — for these segments, a missing value most likely means the reservation was simply not made on behalf of a company, not that a record is missing.
- In contrast, roughly 84% of Corporate bookings do have a valid company ID. A missing company value on a Corporate booking is therefore more likely a genuinely missing piece of information, and is labeled "Unknown Company" rather than "No Company".

This distinction matters because filling every missing value with a single generic label (or the column mode) would either invent a false corporate signal or erase the real difference between "no company was ever involved" and "the company was not recorded".

3.2.2 Decisions Made for Missing Values in Agent

The same business-context approach was applied to agent, this time cross-tabulated against `distribution_channel`:

- When both `market_segment` and `distribution_channel` are Direct, a missing agent value logically means the guest booked directly with the hotel — labeled "Direct Booking".
- For bookings coming through TA/TO or Corporate channels, a missing agent value more likely represents an unknown or unrecorded travel agent — labeled "Unknown Agent".

Why business rules instead of the mode?

Cross-tabulating company and agent against market_segment / distribution_channel showed that nulls cluster structurally in specific segments rather than being randomly scattered. Filling with the column mode would have invented a false corporate or agent signal for bookings where none exists. Encoding the absence itself — as a distinct, meaningful category — preserves the real business information instead of erasing it.

3.2.3 Country and Children

Before filling country, the 452 null records were profiled by `market_segment`, `distribution_channel`, hotel and `customer_type`; no single dominant pattern emerged strongly enough to justify inferring a specific country, so nulls were filled as "Unknown" rather than guessed. The 4 missing children values were filled with 0 and the column cast to an integer type, consistent with treating it as an event/count field where absence of a record most plausibly means zero.

3.3 Handling Invalid or Unusual Values

Once the dataset was de-duplicated and imputed, a second cleaning pass addressed values that are not simply missing, but physically impossible or logically inconsistent. This work is documented as a structured Outlier Analysis (see §3.4) because the notebook treats "invalid values" and "outliers" as two ends of the same screening exercise: every extreme or unusual value is first classified as a data entry error, a statistical-only outlier, or a legitimate business value, and only the first category is corrected or removed.

3.4 Outlier Detection and Treatment

3.4.1 Method: Screen Statistically, Decide Commercially

Most extreme values in a hotel booking system are not errors — they are the long tail of real commercial behavior (a suite in peak August, a 60-night extended stay, a group block booked two years out). The notebook therefore uses the classic Tukey IQR fence only as a screen, never as an automatic decision rule:

```
fence = [Q1 - 1.5*IQR, Q3 + 1.5*IQR]
```

Every flagged column is then classified into one of three buckets, and only the first justifies destructive treatment:

Classification	Meaning	Typical Treatment
Data entry error	Physically impossible or internally contradictory	Remove or correct
Statistical outlier only	Rare but commercially valid	Keep
Legitimate business value	Extreme because the business genuinely works that way	Keep, sometimes flag

3.4.2 Why a Blind IQR Rule Would Have Failed on This Dataset

- Zero-width IQR on discrete columns: for adults, $Q1 = Q3 = 2$, so $IQR = 0$ and the fence collapses to the single point $[2, 2]$. A blind rule would flag 22,899 rows (26.2%) as outliers, including 16,503 perfectly ordinary single-guest bookings.
- Zero-inflation: six columns are more than 90% zeros (previous_cancellations 98%, days_in_waiting_list 99%, babies 99%, required_car_parking_spaces 92%, children 90%). These are event counters where zero is the normal case; any non-zero value sits outside a naive fence by construction, so a blind rule would delete precisely the rows carrying signal — including previous_cancellations, the strongest single behavioral predictor of cancellation in the dataset (67.9% cancel rate vs. 26.7% baseline).

3.4.3 Column-by-Column Decisions

Each numeric column was examined individually against its business context before any treatment was applied. The key decisions:

ADR (Average Daily Rate)

Finding	Classification	Decision
1 record at 5,400 (next highest observed rate is 510, for a 1-night stay in the cheapest room category)	Data entry error	Remove — the record alone drove skewness to 10.92 and kurtosis to 981.6, and the booking was cancelled so removing it costs no real revenue
1 record at -6.38 (negative rate, impossible)	Data entry error	Floor at 0 — the rest of the record is a coherent, valid 10-night stay, so the field is corrected rather than the row deleted
1,778 records at exactly 0 (2.03%)	Legitimate business value	Keep — 91% of the entire Complementary segment is zero-rated (comps, staff stays, loyalty redemptions); these cancel at only 10.4% vs. 27.5% overall, confirming

Finding	Classification	Decision
		they are a real, distinct customer category
2,490 records above the Tukey fence (227)	Statistical outlier only	Keep — p99 is 261 and p99.9 is 335; values form a smooth continuum representing family suites in peak season, the hotel's most valuable business

Party size (adults, children, babies)

Finding	Classification	Decision
13 records with 10–55 adults	Legitimate business value	Keep + flag as <code>is_group_block</code> — every one is <code>customer_type = Group</code> , <code>ADR = 0</code> , Resort Hotel, Portuguese, booked 304–361 days ahead: a tour-operator allocation block, not typos
166 records with zero total guests	Data entry error	Remove — a booking describing nobody staying cannot be a real stay; 135 of 166 also carry zero ADR (no revenue)
219 records with children/babies but no adults	Legitimate business value	Keep + flag as <code>is_child_only_record</code> — all City Hotel, 94% room type B, with a real median ADR of 87: a child-supplement line attached to a parent booking
3 records with 9–10 infants/children	Data entry error	Cap — babies clipped to a maximum of 2, children to a maximum of 3; the stay itself is otherwise valid

Stay length (stays_in_week_nights / stays_in_weekend_nights)

- Zero-night bookings — 651 records (0.74%) → Remove. All 651 have `ADR = 0`; a reservation with zero nights and zero rate produces no room-nights and no revenue, and 622 are marked Check-Out, which is self-contradictory for a zero-night stay.
- Stays over 30 nights — 23 records, max 69 nights → Keep. Mean ADR in the 15–30 night band is 86 and in the 30+ band 49 (against 106 overall), consistent with a genuine extended-stay discount rather than data corruption.

A genuine insight surfaced by this step: cancellation risk rises steadily with stay length and spikes to 54.4% in the 15–30 night band, meaning long, discounted stays are the most fragile bookings in the portfolio — an insight that would have been lost had these rows been dropped as outliers.

Lead time

Kept entirely. Skewness (1.43) and kurtosis (2.13) are normal for a duration variable, the long-lead tail is smooth rather than spiky, and bookings made more than a year ahead cancel at 40.9% versus 27.5% overall — making `lead_time` the single most useful continuous predictor in the dataset.

Zero-inflated behavioral counters

previous_cancellations, previous_bookings_not_canceled, booking_changes, days_in_waiting_list, required_car_parking_spaces and total_of_special_requests were all kept, with one narrow cap: required_car_parking_spaces was capped at 3 (two adjacent records requesting 8 spaces for a 2-adult booking were judged implausible). In every other case, the extreme tail carries real signal — for example, previous_cancellations ≥ 1 is associated with a 68.2% cancellation rate versus 26.9% for guests with none.

3.4.4 Outlier Decision Register and Applied Treatment

Every decision above was consolidated into an auditable decision register before any row was actually dropped. The final treatment removed only 758 rows — 0.87% of the dataset — and every removed row is a record that provably cannot describe a real stay (no guests, no nights, or an impossible rate):

```
drop_no_guest = (adults + children + babies) == 0 drop_no_night = (stays_in_week_nights +
stays_in_weekend_nights) == 0 drop_adr_error = adr > 1000 data_clean = data.loc[~(drop_no_guest |
drop_no_night | drop_adr_error)] # corrective caps -- fix the impossible field, keep the valid
stay data_clean['adr'] = data_clean['adr'].clip(lower=0)
data_clean['babies'] = data_clean['babies'].clip(upper=2)
data_clean['children'] = data_clean['children'].clip(upper=3)
data_clean['required_car_parking_spaces'] =
data_clean['required_car_parking_spaces'].clip(upper=3)
```

Metric	Before	After
Rows	87,396	86,638
ADR skewness	10.92	0.93
ADR kurtosis	981.6	1.58
Cancellation rate	27.49%	27.68% (target balance preserved)

Was a transformation (e.g. log) applied to ADR?

No. ADR's near-total shape correction came from removing one erroneous €5,400 record, not from a mathematical transform. A $\log_{1p}$ transform was tested and rejected: because 1,053 legitimate zero-rated bookings stay pinned at $\log(1+0) = 0$ while the right tail compresses, $\log_{1p}(\text{adr})$ turns a mild right-skew of +0.93 into a severe left-skew of -3.49 — actively worse. ADR is left on its original, interpretable euro scale. The zero-inflated behavioral counters are likewise left untransformed, since no monotone transform fixes a 90%+ zero mass; the correct treatment for those is the binary/categorical encoding built in Feature Engineering.

4. Exploratory Data Analysis (EDA)

4.1 Dataset Overview After Cleaning

- 86,638 bookings × 32 original columns, zero duplicates, zero missing cells — the frame is analysis-ready.
- Class balance: 27.68% cancelled. Imbalanced but workable for descriptive analysis without resampling; it does mean that raw accuracy would be a misleading metric for any future predictive model (a "never cancels" model would score 72%).
- Coverage is not a full three years — 2015 covers July–December and 2017 covers January–August; only 2016 is complete. Any year-over-year comparison must be month-matched.
- High-cardinality identifiers: agent (335 levels), company (349 levels), country (178 levels) — these are keys, not categories, and were not one-hot encoded (see Feature Engineering §5.6).
- reservation_status is a perfect proxy for the target variable and must never be used as a predictor (see §4.5).

4.2 Statistical Analysis

A reusable `describe_plus()` helper extends pandas' `describe()` with skewness, kurtosis and mode, since these shape statistics — not the mean/median alone — are what determine whether a transformation or special encoding is needed.

Column	Mean	Median	Mode	Skewness	Interpretation
lead_time	80	50	0	1.43	A large same-day mass plus a long planning tail; same-day bookings almost never cancel (6.0%)
adr	107	99	75	0.93	Now near-symmetric after outlier treatment; mean ADR is safe to use as a headline metric
adults	—	2	2	20.5 (raw)	74% of bookings are exactly 2 adults; extreme skew driven entirely by the 13 group-block records
total_of_special_requests	—	—	0	1.08	The only near-symmetric behavioral column; usable

Column	Mean	Median	Mode	Skewness	Interpretation
					as a predictor with no transformation
previous_cancellations	—	0	0	34.2	Zero-inflated by design (98% zero); not fixable by transformation — encoded as a ratio/binary in Feature Engineering

4.3 Categorical and Numerical Analysis

Categorical composition

- Channel concentration is the headline structural risk: Online TA alone is 59.2% of all bookings, and travel agents/tour operators (TA/TO) account for 79.2% of distribution overall. Direct bookings are only 13.5%.
- deposit_type is effectively a constant: 98.7% "No Deposit", 1.2% "Non Refund", 0.1% "Refundable" — the hotels take almost no financial guarantee on the average booking, which is the structural reason a 27.7% cancellation rate is possible at all.
- customer_type: 82.4% Transient (independent, unaffiliated) guests; Contract and Group business (typically the stable base) is only 4.2% combined.
- reserved_room_type: Room type A is 64.7% of reservations, leaving thin tails (types L and P have single-digit counts) whose rates are not statistically interpretable.
- A data-quality note on meal: 485 "Undefined" records are all Resort Hotel and cancel at 16.9%, versus SC (Self Catering, 99% City Hotel) at 35.7%. Because they behave like a genuinely distinct product, "Undefined" was kept as its own category rather than merged into SC, contrary to common generic-cleaning advice.
- A data-quality note on country: both "CN" (1,092 records) and "CHN" (815 records) — the ISO-2 and ISO-3 codes for China — coexist, with very different cancellation rates (20.6% vs. 48.1%), suggesting two different source systems rather than a simple duplicate label. This was flagged rather than merged, pending a source-system answer from the business.

Numeric distributions

Histogram + KDE + boxplot profiles were built for the eight decision-relevant numeric variables (lead_time, adr, stays_in_week_nights, stays_in_weekend_nights, adults, total_of_special_requests, booking_changes, previous_cancellations), each annotated with skewness and kurtosis to drive the transformation decisions documented in §3.4.

4.4 Bivariate Analysis — What Predicts a Cancellation

Every categorical field was compared against the 27.7% overall cancellation baseline.

The counter-intuitive deposit finding

Non Refund bookings cancel at 94.7% — the single most surprising number in the dataset, since a non-refundable deposit is supposed to prevent cancellation. Two mechanisms explain it:

- The policy is applied reactively, not as a deterrent: of the 1,037 Non Refund bookings, 982 are Portuguese and mean lead time is 211 days (versus 80 overall) — revenue management is imposing pre-payment terms on exactly the profile it already distrusts.
- A cancelled non-refundable booking is not a lost sale: the guest forfeits the payment, so the revenue is retained (quantified in §8.1 at €237k).

Practical implication: deposit_type must not be read causally. "Require deposits to cut cancellations" does not follow from this table.

Channel reliability

Reliable channels run at roughly half the baseline: Corporate (12.2%), Direct (14.9%), Offline TA/TO (15.0%). Online TA, at 35.6%, is both the least reliable channel and the largest by volume.

Special requests — the cleanest protective signal

Cancellation falls in a near-perfect monotone gradient from 33.5% at zero special requests to 5.6% at five requests. Because requests are recorded at booking time, this is a genuinely usable early-warning field (unlike parking, which is assigned only on arrival — see §4.5).

Lead time — the strongest continuous driver

Cancellation rises monotonically and steeply: 6.0% for same-day bookings to 41.0% beyond a year — a 6.8x spread. Median lead time is 80 days for cancelled bookings versus 38 days for kept ones. The sharpest single jump is between the 1–7 day window (9.7%) and the 8–30 day window (25.5%), where risk nearly triples.

The two hotels are different businesses

Headline ADR (City Hotel €111.66 vs. Resort €100.12) suggests the City Hotel is the stronger performer, but the monthly view shows this is incomplete. The Resort swings from €49.92 in January to €188.90 in August (a 3.8x annual range) and earns most of its year in about ten weeks, while the City Hotel runs a much flatter 85.84–128.36 range (1.5x) with steady year-round demand. Realised revenue per surviving booking is higher at the Resort (€422) than the City Hotel (€328). The City Hotel is a volume-and-occupancy business; the Resort is a yield-management business — they warrant separate pricing and staffing strategies rather than a single "which hotel performs better" comparison.

4.5 Reserved vs. Assigned Room — and a Target-Leakage Warning

Comparing reserved_room_type against assigned_room_type surfaces the most important modelling caveat in the dataset: three fields encode the booking's outcome rather than predicting it, because they only take a meaningful value once the guest has already arrived or the stay has concluded.

Field	Cancellation Rate	Why It Leaks the Target
reservation_status	100% / 0%	"Canceled" and "No-Show" together are exactly the definition of

Field	Cancellation Rate	Why It Leaks the Target
		is_canceled = 1 — a perfect restatement of the target
required_car_parking_spaces > 0	0.00% (0 of 7,290)	Parking is allocated on arrival, so any non-zero value proves the guest actually arrived
room_changed (reserved ≠ assigned)	4.8% vs. 31.6%	Rooms are reassigned at check-in; only 1.9% of cancelled bookings show a change, versus 19.2% of check-outs

Practical consequence

reservation_status, reservation_status_date, required_car_parking_spaces and room_changed must be excluded from any predictive model. Including them would produce near-perfect validation scores that collapse in production, because none of them is known at the time a booking is made. They remain valid for descriptive analysis of what already happened, which is how they are used throughout this documentation.

4.6 Multivariate Analysis — The Hidden Guest-Origin × Channel Interaction

Single-variable rates hide where risk actually concentrates. Portugal is 31% of all bookings and cancels at 36.4% versus 23.8% internationally — but the interaction with market segment is far sharper than that average suggests:

Market Segment	International	Portugal (Domestic)	Ratio
Groups	1.6%	58.8%	37x
Offline TA/TO	0.5%	44.4%	89x
Corporate	2.7%	15.1%	5.6x
Direct	10.8%	19.6%	1.8x
Online TA	33.2%	44.4%	1.3x

Domestic Offline TA/TO bookings cancel at 44.4% while the identical channel sells to international guests at just 0.5% — the same product, an 89-fold difference in reliability. Averaging these together produces the misleading conclusion that "Offline TA/TO is a safe channel at 15.0%"; it is safe internationally and one of the worst channels domestically. This single interaction — affecting roughly 6,700 domestic Groups and Offline TA/TO bookings — is identified as the most actionable finding in the analysis, because it targets a specific, sizeable book of business without disturbing the reliable international base.

4.7 Correlation Analysis

Every numeric field's Pearson correlation with is_canceled is below |0.19| in absolute value, which — read naively — would suggest nothing predicts cancellation. This directly contradicts the 94.7% and 58.8% rates found in the bivariate analysis above, for three reasons documented in the notebook:

- The real drivers are categorical (deposit_type, market_segment, country), and Pearson correlation cannot see categorical effects at all — a correlation heatmap is the wrong instrument for this dataset.

- Relationships are non-linear: `lead_time` shows Pearson 0.184 vs. Spearman 0.225; `previous_cancellations` shows Pearson 0.051 vs. Spearman 0.127 — a 2.5x difference, because a single extreme value of 26 drags the linear fit while the rank correlation captures the real, binned pattern.
- Correlation with a binary target is mechanically bounded by class balance, so at 27.7% positives, even a genuinely strong effect cannot produce a high raw correlation coefficient.

Ranked by Spearman correlation, the numeric predictors order as: `lead_time` (+0.225), `required_car_parking_spaces` (−0.188, flagged as leakage), `adr` (+0.138), `total_of_special_requests` (−0.129), `previous_cancellations` (+0.127), `booking_changes` (−0.124). Multicollinearity is mild — only two pairs exceed $|0.5|$: `stays_in_week_nights` / `stays_in_weekend_nights` (0.547) and `arrival_date_year` / `arrival_date_week_number` (−0.514, an artifact of the incomplete 2015/2017 coverage).

5. Feature Engineering

35 new features were engineered, each justified against a specific finding from the EDA rather than created generically. The full register, including deliberately rejected feature ideas, is documented in the notebook's §3.6.

5.1 Stay & Value Features

Feature	Type	Business Purpose
total_nights	Numeric	Length of stay is the revenue multiplier; risk was found to peak in the 15–30 night band
total_guests	Numeric	Occupancy per booking, needed for per-guest yield analysis
weekend_ratio	Numeric	Separates leisure (weekend-heavy) demand from business (weekday) demand
is_family	Binary	Families showed 34.3% cancellation and the highest ADR (157) — a distinct segment worth isolating
adr_per_guest	Numeric	Raw ADR alone conflates room rate with party size
total_revenue	Numeric	$\text{adr} \times \text{total_nights}$ — the actual money at stake per booking, not just its rate
guest_composition	Categorical	Solo / Couple / Family / Large party / Multi-adult / Child-only — distinct behavioral segments
stay_length_category	Categorical	A usable, human-readable banding of the heavily skewed nights variable

guest_composition immediately separates the portfolio by value and risk: Families pay the highest rate (ADR 157) but cancel at 34.3%; Solo travellers are the cheapest (80) and by far the most reliable (19.4%); Couples are 65% of all bookings at 28.4% cancellation. total_revenue is what makes the revenue-at-risk analysis in §8.1 possible, since cancellation rate alone treats a €75 one-night stay and a €3,000 two-week suite booking as equal losses.

5.2 Temporal Features

Feature	Type	Business Purpose
arrival_date	Datetime	Constructed from the year/month/day parts; also serves as a validation step for date consistency
arrival_quarter	Categorical	Standard quarterly reporting grain

Feature	Type	Business Purpose
booking_season	Categorical	Winter/Spring/Summer/Autumn — the two hotels showed strong, opposite seasonality
arrival_day_of_week	Categorical	Operational staffing signal
is_weekend_arrival	Binary	Flags Friday/Saturday arrivals for staffing and pricing
is_high_season	Binary	Defined data-drivenly as the months with above-average ADR, not assumed from a calendar

Building arrival_date doubles as a validation step: zero invalid dates were produced, confirming the three raw date-part columns are internally consistent. is_high_season captures April–September (the six above-average-ADR months). The resulting seasonal table shows Summer carries both the highest cancellation rate (31.7%) and the highest mean revenue per booking (574), versus Autumn at 23.5% and 322 — peak season concentrates the most money in the least reliable bookings, which is the economic argument for a seasonal deposit policy rather than a flat one.

5.3 Behavioral and Commercial Features

Feature	Type	Business Purpose
total_previous_bookings	Numeric	Combines the two sparse history counters into one
has_booking_history	Binary	Simple flag for any prior guest history at all
previous_cancel_ratio	Numeric	A bounded [0,1] rate of prior cancellation, fixing the extreme skew (34.2) of the raw count
is_repeat_customer	Binary	A broader loyalty definition than the raw is_repeated_guest flag alone
booking_modified	Binary	Correct encoding for the 82%-zero booking_changes counter
was_waitlisted	Binary	Correct encoding for the 99%-zero days_in_waiting_list counter
requests_category	Categorical	Bands the cleanest protective numeric signal (None / Low / Medium / High)
booking_advance_category	Categorical	Lead-time bucket used throughout the bivariate analysis
adr_category	Categorical	Quartile-based Budget / Standard / Premium / Luxury pricing tiers
is_domestic	Binary	Flags Portuguese (PRT) guests — the single strongest interaction effect found in the data

Feature	Type	Business Purpose
booking_source_group	Categorical	Collapses market_segment × distribution_channel (40 sparse combinations) into 6 managed commercial channels

booking_source_group turns two overlapping, high-cardinality columns into six channels a commercial team can actually manage, producing the clearest strategic table in the analysis: Online OTA is 59.2% of bookings, 64.6% of revenue, and cancels at 35.6% — the business's largest revenue source is also its least reliable. previous_cancel_ratio fixes the correlation-collapse problem identified in §4.7, since a ratio is bounded and comparable between a guest with 1 prior booking and one with 72.

5.4 Data-Integrity Flags

These preserve the outlier-analysis findings as reusable columns rather than silent deletions, so any downstream analysis can exclude structurally atypical records on demand without re-deriving the logic.

Feature	Definition	Purpose
is_group_block	adults ≥ 10 and customer_type = Group	Flags tour-operator allocation blocks so per-booking occupancy averages can exclude them
is_child_only_record	adults = 0 and (children + babies) > 0	Flags child-supplement records attached to a parent booking
is_zero_rated	adr = 0	Flags comp/staff/loyalty stays so ADR-based analysis can include or exclude them explicitly

5.5 Composite Cancellation Risk Score

The final and most consequential feature combines the independent signals found throughout the EDA into a single, additive, interpretable score, built to three explicit design constraints:

- Leakage-free — built only from fields known at booking time. No reservation_status, no required_car_parking_spaces, no room_changed.
- Weighted by observed effect size — the three strongest findings (Non Refund deposits at 94.7% cancellation, prior cancellations at 68.2%, and the lead-time gradient) receive the heaviest weights; protective signals (special requests, repeat-customer status, booking modifications) subtract from the score.
- Interpretable — a front-desk manager can read exactly why a booking scored what it did, unlike a fitted black-box model.

The notebook is explicit that this is a transparent, hand-weighted scorecard, not a fitted statistical model — the validation below is a consistency check that the component signals combine sensibly, not an out-of-sample performance claim.

```
score = 0 score += lead_time_bucket_weight(0..4) # 6.8x monotone gradient score += 3
if deposit_type == 'Non Refund' score += 3 if previous_cancellations >= 1 score += 1 if
```

```
market_segment == 'Online TA' score += 1 if is_domestic score += 1 if customer_type == 'Transient'
score += 1 if total_of_special_requests == 0 score -= 1 if total_of_special_requests >= 2 score -=
booking_modified score -= is_repeat_customer risk_score = max(score, 0)
```

Risk Tier	Bookings	Share	Cancellation Rate	Revenue at Risk
Low	7,043	8.1%	3.6%	—
Moderate	31,265	36.1%	12.6%	—
High	35,052	40.5%	31.5%	—
Very High	13,278	15.3%	66.0%	€6.8M

The tiers separate cleanly and monotonically, producing an 18.2x lift from the Low to the Very High tier with no inversions in the underlying per-score curve. Operationally, the Very High tier is small enough (15.3% of bookings) to manage by hand — deposit requirements, reconfirmation calls, or planned overbooking — without disturbing the 44% of bookings sitting in the reliably-converting Low/Moderate tiers.

5.6 Rejected Features and the Leakage Exclusion List

Recording what was deliberately not built is treated as part of the analytical output:

- `is_cancelled_before` (from `previous_cancellations > 0`) — already covered by `previous_cancel_ratio` and `has_booking_history`; a third encoding of the same column would add collinearity, not information.
- `revenue_lost` (total_revenue where cancelled) — a direct function of the target variable; belongs in reporting only, never in a feature set.
- `room_upgraded` / `room_downgraded` — tempting from the reserved-vs-assigned heatmap, but room-type letters are anonymized with no published quality ordering, so "upgrade" cannot be objectively defined.
- One-hot encoding of country, agent, company — 862 combined levels would produce a matrix wider than it is tall; `is_domestic` captures the effect that actually matters, and the rest would need frequency- or target-encoding if ever required.
- Any transformation of `adr` — the `log1p` test in §3.4 showed it makes the distribution materially worse.

Leakage exclusion list (must be excluded from any predictive model)

reservation_status, reservation_status_date, required_car_parking_spaces, room_changed — all four encode information that is only known at or after arrival, and would leak the outcome directly.

Final engineered dataset: 86,638 bookings × 67 columns, zero missing values, zero duplicates.

6. Data Visualization

Visualizations were built throughout the notebook using a consistent visual language: a fixed color palette (navy, teal, coral, gold, grey), a shared seaborn theme, and reusable helper functions (`bar_with_labels`, `profile_numeric`, `cancel_rate_by`) so that every chart reads the same way. The main categories of Python visualization used are described below, followed by the Power BI dashboard (§7), which is the primary interactive deliverable.

6.1 Distribution Profiling

Histogram + KDE + boxplot combinations (`profile_numeric()`) were used for key numeric fields such as `adr`, `lead_time`, and `total_nights`, both before and after outlier treatment. These made it possible to see, visually, that removing a single erroneous ADR record corrected the distribution's shape (skew 10.92 → 0.93) without needing a mathematical transform.

6.2 Categorical Composition Charts

Bar charts with percentage labels were used for `hotel`, `market_segment`, `distribution_channel`, `customer_type`, `deposit_type`, `meal`, `reserved_room_type` and `arrival_date_month`, making channel concentration (Online TA = 59.2% of bookings) and the near-constant nature of `deposit_type` immediately visible.

6.3 Cancellation-Rate Bar Charts (Bivariate)

A reusable `cancel_rate_by()` function produced labelled bar charts for every categorical field against the 27.7% baseline, colored red (worse than baseline) or teal (better than baseline), with booking volumes (`n=`) printed inside each bar so small, statistically unreliable categories are visible at a glance.

6.4 Trend and Seasonality Charts

Line charts of mean ADR by month per hotel revealed the two properties' opposite seasonal shapes (Resort 3.8x annual ADR range vs. City Hotel 1.5x), and a dual-axis chart (bars for booking volume, line for cancellation rate) showed that demand and cancellation risk rise together, peaking in August.

6.5 Heatmaps (Multivariate)

Cross-tabulated heatmaps were used for: Hotel × Market Segment cancellation rate, Lead Time × Deposit Type cancellation rate, Market Segment × Guest Origin cancellation rate (which surfaces the domestic-channel interaction), Market Segment × Customer Type mean ADR, and Reserved Room Type × Assigned Room Type booking counts.

6.6 Correlation Heatmap and Driver Ranking

A masked lower-triangle Pearson correlation heatmap across all numeric fields, paired with a horizontal bar chart ranking each field's correlation with `is_canceled` — used together with a Pearson-vs-Spearman comparison table to demonstrate why raw linear correlation understates several of the dataset's real drivers.

6.7 Risk-Tier and Revenue-Exposure Charts

A stacked horizontal bar chart broke down realised versus lost revenue by `booking_source_group`, quantifying that Online OTA alone accounts for 83% of all revenue at risk. A ranked horizontal bar chart of cancellation-rate "gaps" (high-risk group vs. comparison group, in percentage points) visualized the ranked driver table used in the recommendations.

7. Power BI Dashboard

The Python analysis feeds a 4-page interactive Power BI report, filterable by hotel, month, customer type and country, using a consistent sage/forest color palette and consistent card-style KPIs across all four pages so that filters read the same way on every view.

7.1 Page 1 — Executive Overview

The landing page presents five headline KPI cards summarizing the entire cleaned dataset:

KPI	Value
Total bookings	87K
Cancellation rate	27.5%
Total revenue	€34.46M
Average ADR	106.34
Average stay (nights)	3.63

7.2 Page 2 — Customer Analysis

Breaks down the guest base by market segment, meal preference, distribution channel, and repeat-guest rate (3.9%). This page is the primary lens for understanding who is booking and through which channel.

7.3 Page 3 — Cancellation Analysis

Cancellation rate sliced by lead time, deposit type and market segment, tracked month by month — the operational page for monitoring the drivers quantified in §4.4–§4.6.

7.4 Page 4 — Revenue & Room Analysis

ADR trends over time, revenue broken down by market segment, and room-type performance compared across the two hotels.

7.5 Filters and Slicers

All four pages share a common set of slicers — hotel, month, customer type and country — so any page can be sliced consistently to the same subset of bookings without losing visual continuity.

7.6 Main Business Insights Highlighted by the Dashboard

- The overall 27.5% cancellation rate and €34.46M revenue base give stakeholders an immediate, governed reference point that matches the cleaned Python analysis exactly (no re-derivation drift between the notebook and the dashboard).
- The Cancellation Analysis page lets a non-technical user reproduce the key EDA finding — that cancellation risk rises sharply with lead time and varies enormously by deposit type and market segment — without needing to read the notebook.

- The Revenue & Room page supports the yield-management vs. volume-and-occupancy distinction between the Resort and City Hotel identified in the Python analysis.
- Because the dashboard is filterable by country, a business user can directly inspect the domestic (Portugal) vs. international cancellation gap that the notebook identifies as the single most actionable finding.

8. Key Findings and Business Problems

8.1 Revenue at Risk — Reframing the Cancellation Problem

Cancellation rate alone is the wrong headline metric, because it weights a €75 one-night stay identically to a €3,000 two-week suite booking. Of €34.4M in total booked room revenue, €11.5M (33.3%) is attached to cancelled bookings. Two corrections matter:

- €237k of that is non-refundable and therefore retained, not truly lost — net exposure is €11.24M (32.6%).
- Revenue exposure (33.3%) exceeds the raw booking cancellation rate (27.7%) — cancelled bookings are systematically more valuable than kept ones, because high-ADR, long-stay, long-lead bookings are the ones most likely to fail.

Online OTA alone accounts for €9.29M of the €11.24M net exposure — 83% of all revenue risk — despite being 59% of bookings and 65% of revenue. Every other channel loses between 8% and 29% of its booked revenue; OTA loses 42%. This reframes the strategic problem from "reduce cancellations across the business" to "the OTA channel is simultaneously the largest revenue source and the origin of five-sixths of the revenue risk."

8.2 Ranked Drivers of Cancellation

Rank	Driver	High-Risk Group	Comparison Group	Gap
1	Non-refundable deposit	94.7%	26.9%	+67.8 pp
2	Domestic group booking	58.8%	1.6%	+57.1 pp
3	Prior cancellation on file	68.2%	26.9%	+41.3 pp
4	Lead time > 180 days	39.8%	8.5%	+31.3 pp
5	First-time guest	28.6%	7.6%	+21.1 pp
6	Online OTA channel	35.5%	14.8%	+20.7 pp

This ranking must be read with the direction of causation in mind, and it differs by row:

- Rank 1 is not causal — Non Refund terms are applied to bookings already judged risky; requiring deposits more widely would not reproduce this gap.
- Ranks 2–4 are genuine, actionable risk markers — guest origin, behavioral history and booking horizon are all known at booking time and are not consequences of the outcome.
- Ranks 5–6 are structural — they describe which book of business is fragile, not a lever that can be pulled directly.

The most useful finding is rank 2, because it is both a large effect and precisely targetable: domestic Groups cancel at 58.8% against 1.6% for the same segment sold internationally — a 37x difference — covering roughly 6,700 bookings where intervention would not disturb the reliable international base.

8.3 Answers to the Core Business Questions

Which customers cancel the most?

Domestic (Portuguese) guests booking through Groups (58.8%) or Offline TA/TO (44.4%) channels, and any guest with a prior cancellation on file (68.2%). Overall, domestic guests cancel at 36.4% versus 23.8% international. By party type, families cancel at 34.3% and solo travellers at 19.4%.

Which booking channels are the most reliable?

Corporate/Business (12.9%), Direct (14.8%) and Offline Travel Trade (15.0%) — but the last only internationally; its domestic book runs at 44.4%. Online OTA at 35.6% is the least reliable and the largest channel.

Which customer types generate the highest ADR?

Transient guests, at a 111 mean ADR, ahead of Contract (93), Transient-Party (88) and Group (86). Crossed with channel, Online TA + Transient is the top-yielding cell (119). The commercial tension is that the highest-ADR segments are also the highest-cancelling ones.

Which hotel performs better?

Neither, unqualified — they are different businesses. The City Hotel has higher headline ADR (111.66 vs. 100.12), 61% of volume and steady year-round demand, but cancels more (30.2% vs. 23.7%). The Resort has extreme seasonality (ADR 49.92 in January to 188.90 in August), cancels less, and earns more per realised booking (422 vs. 328). The City Hotel is a volume-and-occupancy business; the Resort is a yield-management business.

Which countries have the highest cancellation rate?

Among countries with 500+ bookings: China (CHN) 48.1%, Russia 39.9%, Brazil 36.5%, Portugal 36.4%, Italy 35.1%. The most reliable are Austria 17.9%, Netherlands 18.3%, Great Britain 19.1%, Germany 19.6% and France 19.6%. Portugal matters most in absolute terms because it is 31% of all bookings — the others are individually small. (Note the CN/CHN coding issue flagged in §4.3 before acting on the China figure.)

8.4 Main Problems Identified and Likely Reasons

Problem	Likely Reason
Channel concentration in the least reliable channel	Online TA is both the largest booking source (59.2%) and the highest-cancelling channel (35.6%), so growth in volume has come with growth in risk
Domestic Groups / Offline TA/TO cancel 37–89x more than the identical international product	Likely speculative domestic holding — local guests and operators block rooms with no deposit and no sunk travel cost, then decide later; international guests already have flights booked
A flat, near-absent deposit policy (98.7% No Deposit)	The hotels take almost no financial guarantee on the average booking, which structurally enables a 27.7% cancellation rate
Peak season carries peak revenue risk	August is simultaneously the highest-volume, highest-rate and among the highest-cancelling months, so a flat

Problem	Likely Reason
	deposit policy under-protects exactly the inventory worth the most
Long stays (15–30 nights) cancel at over 54%	These are heavily discounted, long-horizon bookings that tie up inventory for weeks while carrying elevated cancellation risk
Two data-quality issues distort reporting if left unresolved	CN/CHN dual country coding for China and a Resort-only "meal = Undefined" category both need a source-system answer from the business

8.5 Business Recommendations

- **1. Fix the domestic Groups / Offline TA/TO leak first.** Roughly 6,700 bookings cancel at 44–59% against international rates of 0.5–1.6% on the identical product. Require a partial deposit or a 72-hour reconfirmation for domestic group and offline-trade bookings only — the highest effect-per-unit-of-disruption action available, and it leaves the reliable international base untouched.
- **2. Treat the Very High risk tier as a managed book.** 13,278 bookings, 66% cancellation, €6.8M attached — small enough for active management via reconfirmation contact, deposit terms, or planned overbooking, using the transparent risk scorecard so front-desk staff can act on it directly.
- **3. Shift OTA volume toward Direct.** Direct loses 19% of booked revenue versus 42% for OTA, at a comparable ADR (111 vs. 119). Moving 10% of OTA volume to Direct is worth roughly €500k in realised revenue, before any commission savings.
- **4. Solicit preferences from long-lead bookers.** Special requests cut cancellation from 44.5% to 24.0% at a 91–180 day horizon, independent of lead time. A pre-arrival preference email is close to free and targets exactly the population that fails most.
- **5. Make deposit policy seasonal, not flat.** Summer carries both the highest cancellation rate (31.7%) and the highest revenue per booking (574). A flat 98.7% no-deposit policy under-protects exactly the inventory that is worth the most.
- **6. Re-price or restructure long stays.** The 15–30 night band cancels at over 54% while occupying inventory for weeks at a discounted rate. Either a deposit or a shorter free-cancellation window is warranted.
- **7. Resolve two data-quality items before they distort reporting.** The CN/CHN country-coding split and the Resort-only meal = Undefined category need a source-system answer from the business before either is combined or separated.

9. Project Workflow

The complete process from raw data to final dashboard followed four structured stages, executed in a single documented Python notebook and delivered as both an analytical record and a Power BI report.

Stage 0 — Data Integrity Setup

Loaded the raw 119,390-row CSV, removed 31,994 exact duplicates, and applied business-driven imputation for company, agent, country and children. Caught and corrected an out-of-order notebook execution that had silently re-introduced duplicates, re-validated the frame with an idempotent integrity check, and locked in the 87,396-row analysis baseline.

Stage 1 — Outlier Analysis

Screened every numeric column with a Tukey IQR fence used strictly as a diagnostic, then classified each flagged pattern as a data-entry error, a statistical-only outlier, or a legitimate business value. Built and applied a conservative, fully-justified treatment that removed only 758 rows (0.87%) and corrected two impossible ADR values, producing a clean 86,638-row frame with no need for a mathematical transform.

Stage 2 — Exploratory Data Analysis

Ran a structural audit of the cleaned frame, then univariate, bivariate, multivariate and correlation analysis to answer the project's core business questions — identifying the deposit-type paradox, the lead-time gradient, the protective effect of special requests, the domestic-guest × channel interaction, and three target-leaking fields to exclude from any future model.

Stage 3 — Feature Engineering

Built 35 new features across stay/value, temporal, behavioral/commercial, data-integrity-flag and composite-risk-score categories, each tied to a specific EDA finding. Documented rejected feature ideas and formalised a leakage exclusion list, arriving at a final analysis-ready dataset of 86,638 bookings × 67 columns.

Stage 4 — Business Insights

Quantified revenue at risk (€11.24M net exposure, 83% traced to the Online OTA channel), ranked the drivers of cancellation with explicit causal caveats, answered the project's core business questions directly from the cleaned and engineered data, and produced seven prioritised, quantified recommendations.

Stage 5 — Power BI Delivery

Published the cleaned and engineered dataset into a 4-page Power BI report (Executive, Customer, Cancellation, Revenue & Room), with consistent KPI cards, a shared color palette, and shared filters (hotel, month, customer type, country) so business stakeholders can explore the same findings interactively without needing to read the notebook.

Summary of Outputs by Stage

Stage	Key Output
0. Integrity	Diagnosed and repaired an out-of-order execution issue that would otherwise have left a 9.5 percentage-point error in the headline cancellation rate
1. Outliers	All numeric columns classified; 758 rows (0.87%) removed, all provably non-stays; ADR skew reduced from 10.92 to 0.93 with no transformation
2. EDA	Univariate, bivariate, multivariate and correlation analysis; identified the domestic-channel interaction and three target-leaking fields
3. Features	31–35 engineered features (per notebook register), each justified; rejected features documented; leakage exclusion list defined
4. Insights	€11.24M net revenue exposure quantified, 83% traced to one channel; ranked drivers with causal caveats; 7 recommendations
5. Dashboard	4-page Power BI report with shared KPIs and filters, published for ongoing self-serve monitoring

Stated Limitations

- The date window is not three full years — 2015 covers July–December and 2017 covers January–August; only 2016 is complete. All year-over-year comparisons must be month-matched, and no genuine multi-year trend can be read from three partial years.
- 31,994 exact duplicates were removed on the assumption that identical rows are duplicate records. The dataset has no booking ID, so it is possible some are genuinely distinct same-day group bookings; this assumption materially affects every rate in the analysis (27.7% de-duplicated vs. 37.0% raw) and should be confirmed against the source system.
- agent and company remain high-cardinality identifiers (335 and 349 levels) and were not encoded; the "Unknown Agent" / "Unknown Company" buckets mix genuinely absent values with unrecorded ones.
- The composite risk score is a hand-weighted scorecard, not a fitted model. Its weights come from effect sizes measured on the same data used to validate it, so the 18.2x lift is an in-sample consistency check, not an out-of-sample performance estimate — a hold-out validation would be required before any operational use.
- All findings are associational, not causal. The deposit finding is the clearest illustration of why no row in the driver table should be read as causal without a controlled test.

10. AI Assistance

AI tools were used as a supporting assistant throughout this project, in the same way a research assistant, rubber-duck reviewer, or documentation aid might be used. Specifically, AI assistance contributed to:

- Brainstorming analytical approaches and framing the core business questions to investigate.
- Helping to understand the structure and quirks of the dataset during initial exploration.
- Suggesting candidate outlier-treatment and missing-value strategies, which were then evaluated against the actual data before being adopted or rejected.
- Improving and reviewing Python code for clarity, reusability (e.g. the shared helper functions) and readability.
- Contributing ideas during feature engineering, which were validated against EDA findings before being implemented.
- Assisting with the structure, clarity and presentation of this documentation and the accompanying project slide deck.

AI as a supporting tool, not a replacement for the analytical work

Every cleaning rule, outlier decision, feature definition, statistical interpretation, and business recommendation in this project was reviewed, validated against the actual data, and implemented as part of the documented project workflow. AI assistance supported the process — it did not perform the analysis, make the final decisions, or generate the findings independently. The decision registers, cross-tabulations, and evidence tables throughout this document exist specifically so that every conclusion can be traced back to and verified against the underlying data.

Conclusion

This project demonstrates the full analyst workflow end to end: business-driven data cleaning (rather than blind, generic imputation), structured multi-layer EDA, justified feature engineering with an explicit rejected-features and leakage-exclusion record, and the translation of findings into both quantified, prioritised business recommendations and a self-serve Power BI dashboard for non-technical stakeholders.

Metric	Result
Final dataset	86,638 rows × 67 columns
Net revenue exposure quantified	€11.24M
Risk-tier lift from the composite score	18.2x (Low to Very High)
Dashboard delivered	4-page Power BI report

Hamza Ahmed Bayoumi — Data Analysis Track, Digital Egypt Pioneers Initiative (DEPI)